



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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How Virtualization Enabled the Growth of Cloud Service Providers

Course Name:CSA1522-Cloud Computing and Big Data Analytics

CO1-AT2-Technical Presentation

Guided by:
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Presented by:
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SIMATS ENGINEERING
ASSESSMENT TOOL 2: Technical Presentation

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Divyasree .M
Register Number	192524098

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I Divyasree |M/192524098 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks**Technical Presentation Topics**

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

1. Cloud computing characteristics and benefits for startup ecosystems.
2. How virtualization enabled the growth of cloud service providers.
3. A comparative presentation on IaaS, PaaS, and SaaS for an educational institution.
4. Web services in cloud platforms: SOAP vs REST for enterprise integration.
5. Why XaaS is becoming a dominant consumption model in modern IT.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Introduction to Virtualization

- Virtualization is the technology that allows multiple virtual machines (VMs) to run on a single physical computer.

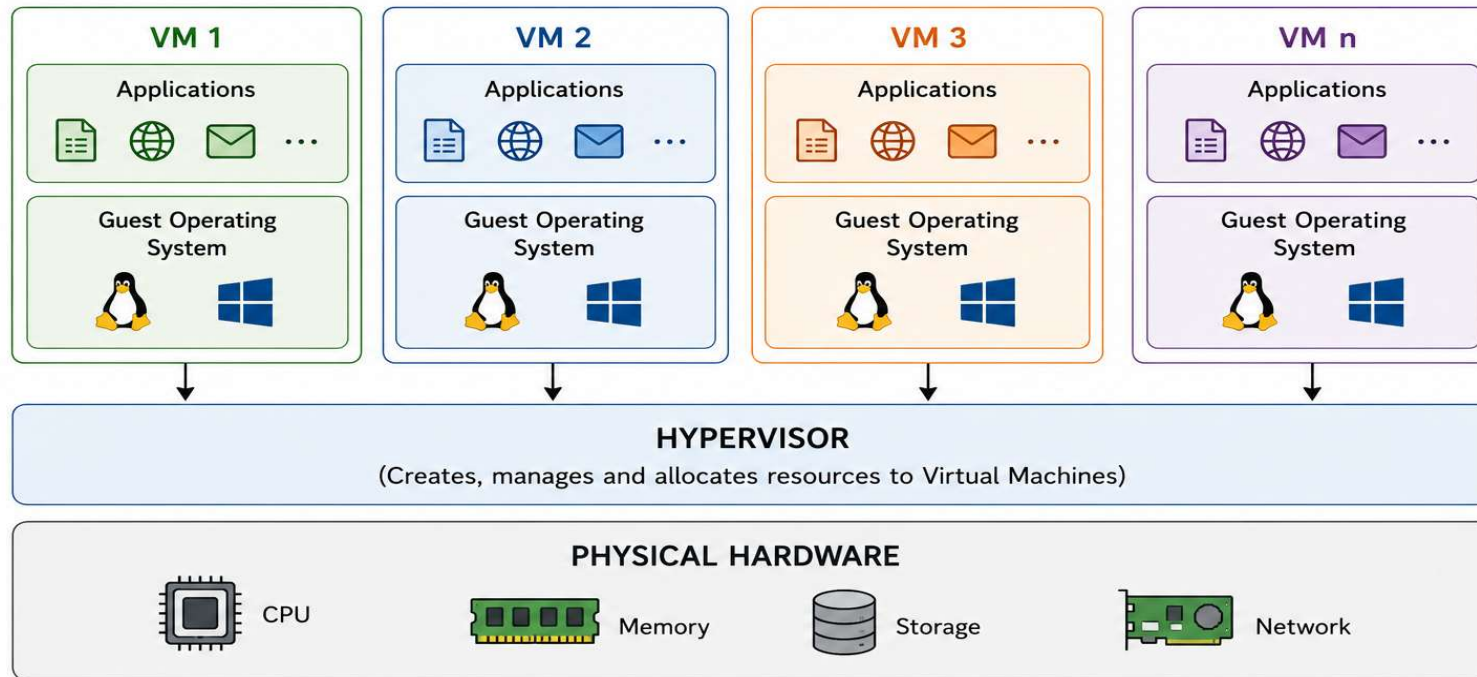
Key Idea

- One physical server can host many virtual servers.
- Each VM has its own operating system and applications.
- Resources such as CPU, memory, and storage are shared efficiently.

Benefits

- Better resource utilization
- Reduced hardware costs
- Improved flexibility
- Easier management

Architecture of Virtualization



Key Idea: The hypervisor sits between the physical hardware and virtual machines, allowing multiple isolated VMs to share the same physical resources efficiently.

Role of Virtualization in Cloud Computing

Virtualization became the backbone of cloud computing because it enables:

- **Resource Pooling**-Multiple users share the same infrastructure.
- **Scalability**-Resources can be increased or decreased on demand.
- **Isolation**-Each virtual machine operates independently.
- **Rapid Provisioning**-New servers can be created within minutes.
- **Cost Efficiency**-Maximizes hardware utilization and reduces operational costs.

How Virtualization Enabled Cloud Service Providers

Before Virtualization:

One application per server
Low hardware utilization
High infrastructure cost



After Virtualization:

Multiple virtual servers per machine
Higher utilization
Lower operational costs
Faster deployment



Impact on Cloud Providers

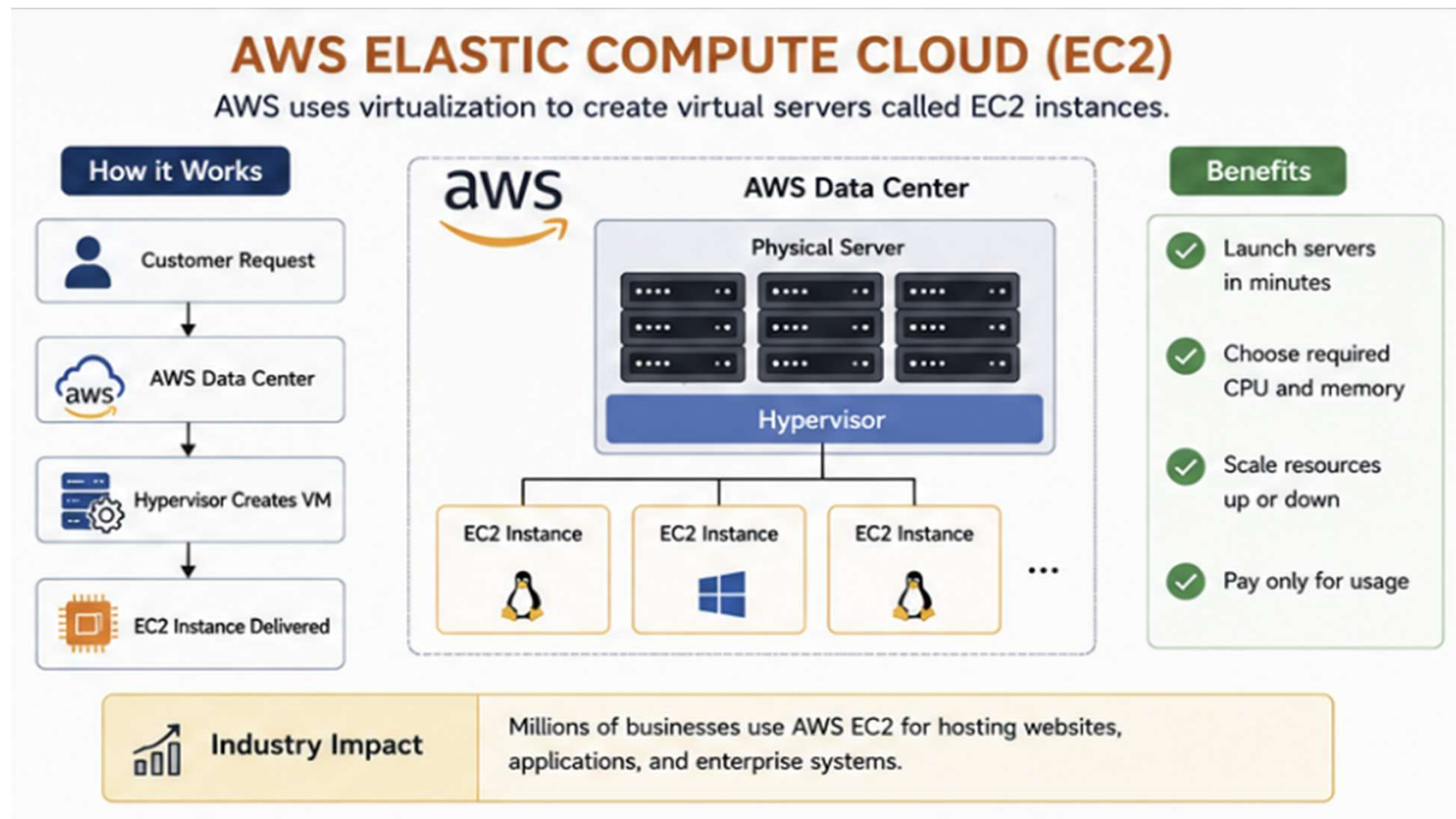
Large-scale data centers became practical.

Customers could rent virtual resources instantly.

Pay-as-you-go model became possible.

Massive scalability was achieved.

Real-World Example – Amazon Web Services (AWS)



Conclusion

- Virtualization allows multiple virtual machines to run on a single physical server.
- It improves resource utilization and reduces costs. Cloud providers use virtualization to deliver scalable and flexible services.
- Technologies such as AWS EC2, Azure Virtual Machines, and Google Compute Engine rely heavily on virtualization.